[PROPERTIES OF SOIL]

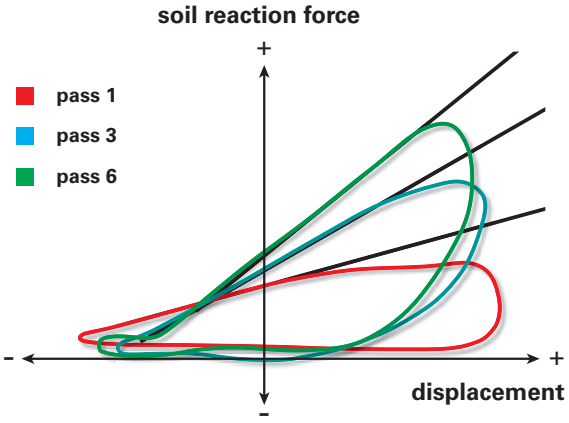
Engineers typically use a number of terms when defining the characteristics and properties of various soils. Understanding these terms is essential to understanding soil compaction principles and techniques.



Load-Bearing Strength is the critical property of a road structure. Simply stated, it is the capacity of a structure to support the load that it carries. Assessment of load-bearing strength is normally achieved by proof rolling with a loaded dump truck and observing rutting or by plate load tests. In road construction, other properties like modulus, stiffness and density are typically used to provide a working target for load-bearing strength.

Stiffness is the ability of a certain shape of material to resist deflection under load and is calculated as the ratio of stress divided by displacement. Unlike elastic modulus, it is not a quality of the soil material itself. Stiffness is a quality of a certain quantity, shape and composition of a soil material—how much that shape of material deflects under load. For that reason, stiffness has become recognized as a valid way to estimate the load-bearing strength of a soil.

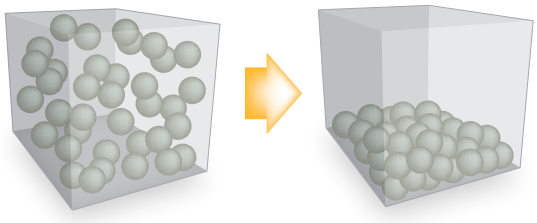
STIFFNESS



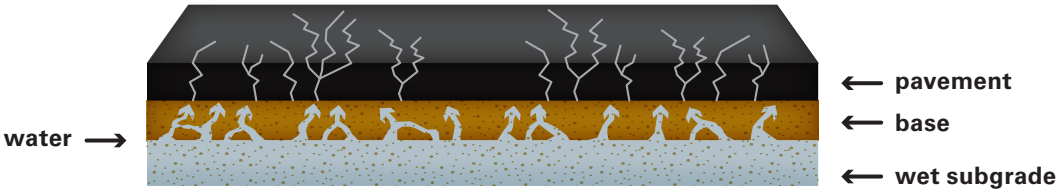
The steeper the slope of the curve, the stiffer the soil.

Density is a calculation of the mass of a material divided by the volume it occupies. Maximum density is the minimum volume a mass of a particular material can occupy. This would be a state with no voids, only a completely solid mass. Soil materials are made denser by compacting them from one volume into a smaller volume. Density has traditionally been the standard property by which engineers estimate load-bearing strength; however, because high density does not correlate to deflection and can cause certain materials to become brittle or degrade, the prominence of density as the primary assessor of load bearing strength has diminished. Regardless, it is still a necessary and trustworthy property to use in making certain assumptions about the support capacity of the road.

DENSITY

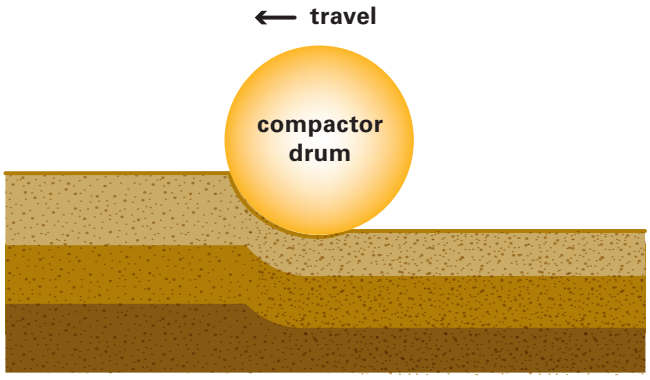


CAPILLARITY



Capillarity is the ability of a soil to force water upward or laterally. A desirable characteristic for base material used as a layer between the subgrade and the pavement of a roadway is to act as a capillary barrier preventing upward capillary movement of water from the subgrade. Granular base also allows water to drain out of the subgrade. Capillary water is held in small pores or voids in the soil. It is considered free water, but it can only be removed by lowering the water table, heavy sustained loading or evaporation. Without a capillary barrier in the base, trapped water would soften and expand the subgrade, resulting in an inadequately supported surface and premature deterioration of the roadway.

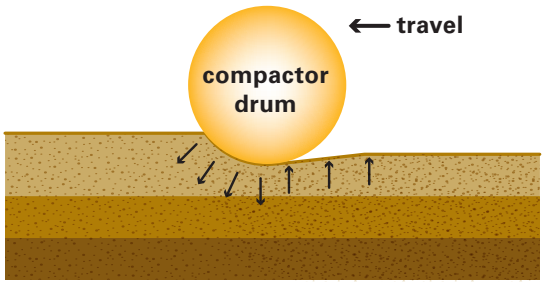
COMPRESSIBILITY



Compressibility is the rate of the reduction in volume of a soil when a force is applied to it. Soils with high compressibility have particles that easily reorient themselves to reduce the space available for air or water voids. In wet conditions, clay soils usually have higher compressibility than granular soils. But they have less permeability, which makes clay soils very slow to drain and compress. When loads are applied rapidly, for example under moving wheel loads, water pressure builds up in fine-grained soils leading to increased compressibility.

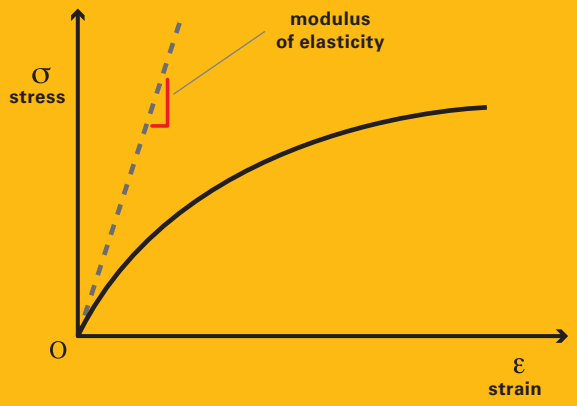
ELASTICITY

Elasticity is the tendency of a soil to deform and return to its original, or near original, shape after a compressive load is removed. Elasticity can be a desirable characteristic for soils, such as for support of fluctuating loads without accumulation of permanent deformation. However, roads with highly elastic bases or subgrades can perform poorly if the modulus of elasticity is too low, resulting in high strains within in the paving layers. Chemical and mechanical stabilization are often used to control the elastic behavior of soils and base. Organic soils have very high elasticity but low modulus of elasticity.

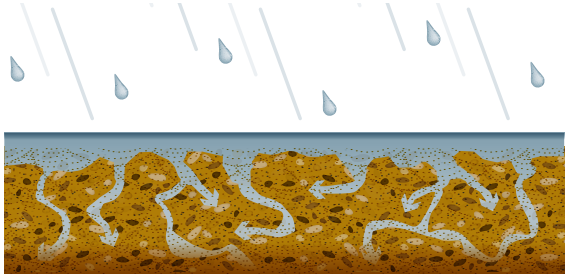


Modulus of Elasticity:

This is a calculation of the ratio of applied stress to the accompanying strain of a soil material. It is considered a quality of the particular soil sample being tested, and can vary with the changing composition of a soil material. The modulus is often used to provide an indication of the load-bearing strength of a soil material. Pavement layer thickness is typically based on an assessment of the underlying modulus of elasticity.

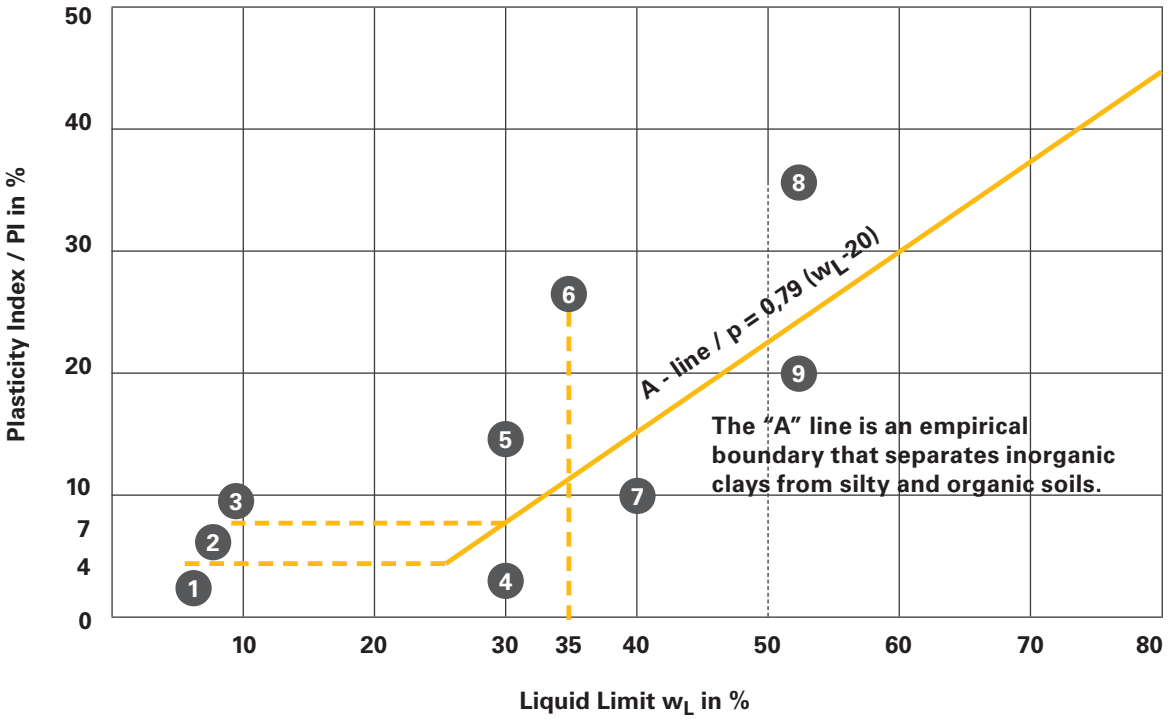


PERMEABILITY



Permeability is the ease with which water flows through a soil. This is not the same as capillarity, which is a soil's ability to absorb water. Soil texture, gradation and the degree of compaction influence a soil's permeability. Permeability is the most variable soil parameter with values varying more than 10 orders of magnitude. Usually, coarse-grained soils are more permeable than fine-grained soils because coarse-grained soils have larger voids between their particles.

PLASTICITY



1. Sand-silt mixtures
2. Intermediate range
3. Sand-clay mixtures
4. Light plastic silts
5. Light plastic clays
6. Medium plastic clays
7. Silts with organic admixtures and organogenic silts and medium plastic silts
8. Distinct plastic clays
9. Clays with organic admixtures and distinct compressible silts

Plasticity refers to the degree of cohesiveness and deformable nature of a soil. The measure of plasticity is expressed as the Plasticity Index (PI). Many clay soils have a high PI value, are quite compressible and have a high degree of cohesion.

A soil with a zero PI is cohesionless or non-plastic. The moisture content of a soil also affects its PI.

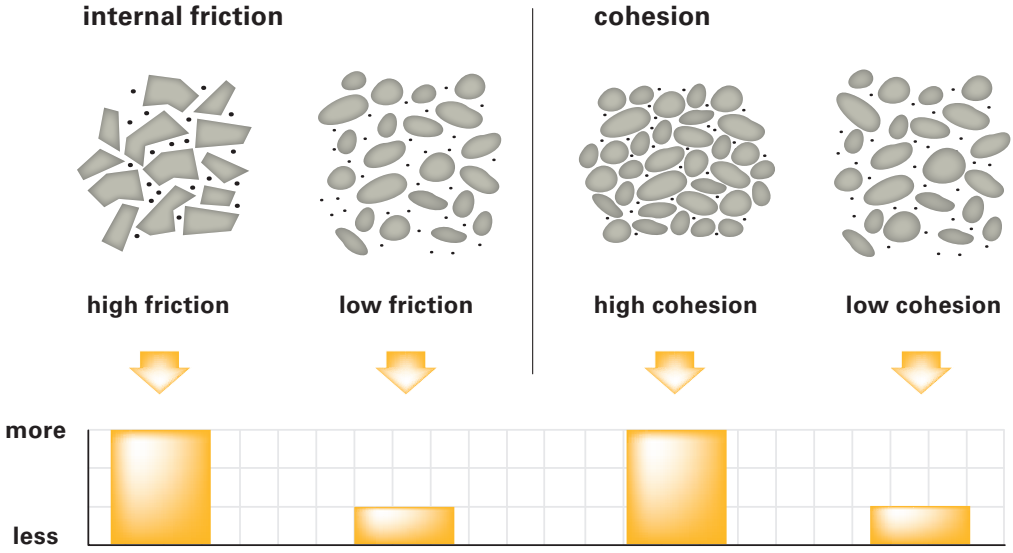
SETTLEMENT



settlement

Settlement is the process of decreasing surface elevation due to the consolidation of fill material. Settlement often is the result of inadequate compaction. Poorly compacted soil particles will, in time, naturally reorient themselves and reduce the space available for air or water. The result is settling, which is directly related to the reduction in the volume of voids.

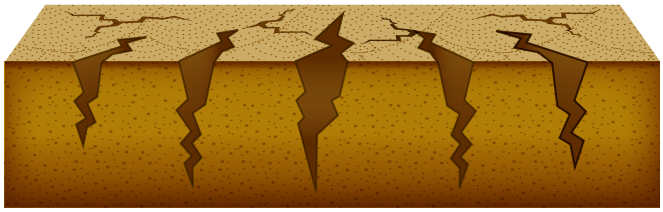
SHEAR RESISTANCE DEPENDS ON ...



Compaction force required to overcome shear resistance

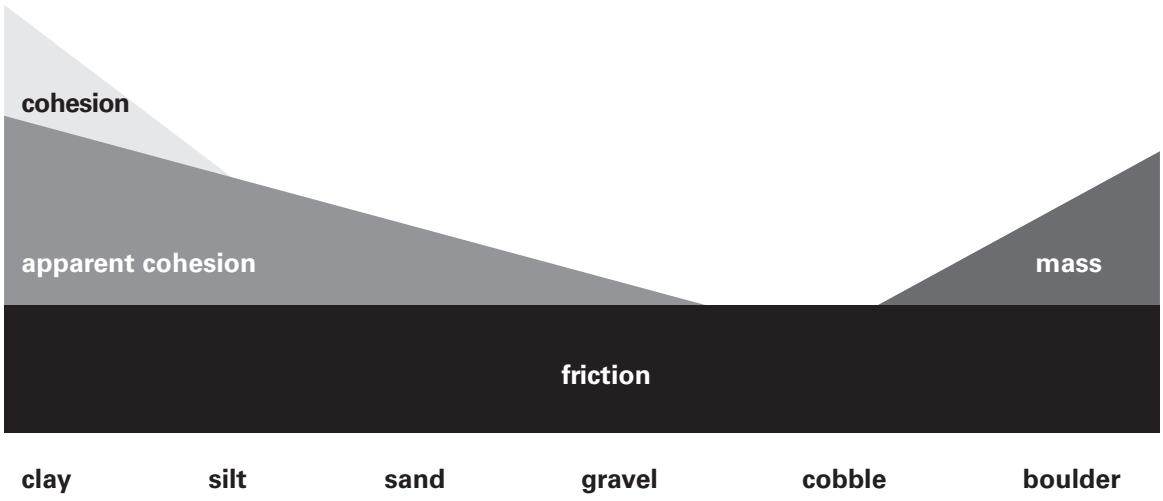
Shear Resistance is the resistance the soil particles have to sliding across each other when a force is applied—for example, vibration or compaction force. The shearing strength of a soil is the result of internal friction (resistance to sliding over each other) and cohesion (attraction to each other). Irregularly shaped particles have higher shear resistance than smooth shaped particles. The greater the shear resistance, the more compactive force is required to achieve density.

SHRINKAGE



Visible Shrinkage or Swelling is an indication that the soil is fine-grained, such as clay. The cycle of shrinking and swelling is a result of the release and build-up of moisture within the soil. This type of soil provides a poor foundation since constant changes in volume can cause structural failure in buildings or pavements dependent on stable support.

COMPACTABILITY



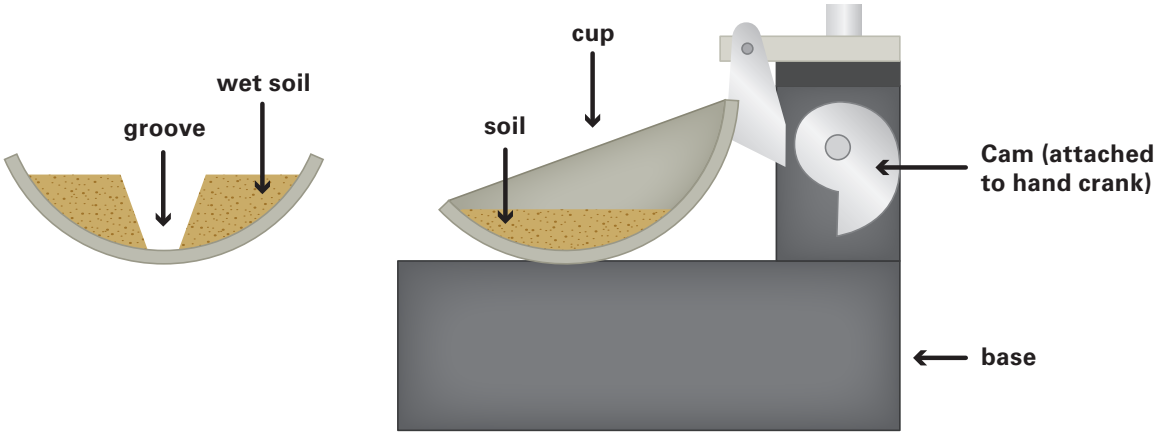
Compactability: During the process of changing a soil from a loose state to a dense state, the ease or rate of compaction is often termed the compactibility. Compactibility can be quantified as the ratio of the difference between the final density minus the initial density divided by the initial density. The higher the compactibility ratio, the easier or more rapid the density will change for the applied compaction effort. Factors that affect compactibility include the soil gradation (well-graded soils tend to have higher compactibility than gap-graded soils), moisture content, shear strength (resistance to deformation), compaction energy and method. Understanding the factors that contribute to increased compactibility will ensure proper equipment selection and efficient compaction operations.

[SOIL LIMITS]

The extent to which moisture content affects the compactability of a cohesive (clay) soil can best be understood by examining soil limits.

Albert Atterberg, a Swedish chemist, first developed certain limits of soil consistency: Liquid Limit, Plastic Limit, Plasticity Index and Shrinkage Limit. Sometimes called the Atterberg Limits, these are the basis for differentiation between highly plastic, slightly plastic and non-plastic materials.

LIQUID LIMIT (LL) TEST



simple liquid limit testing device

Liquid Limit (LL)
The moisture content at which a soil passes from a plastic state to a liquid state is the Liquid Limit. This means that there is enough moisture in the soil to overcome internal friction and cohesion.

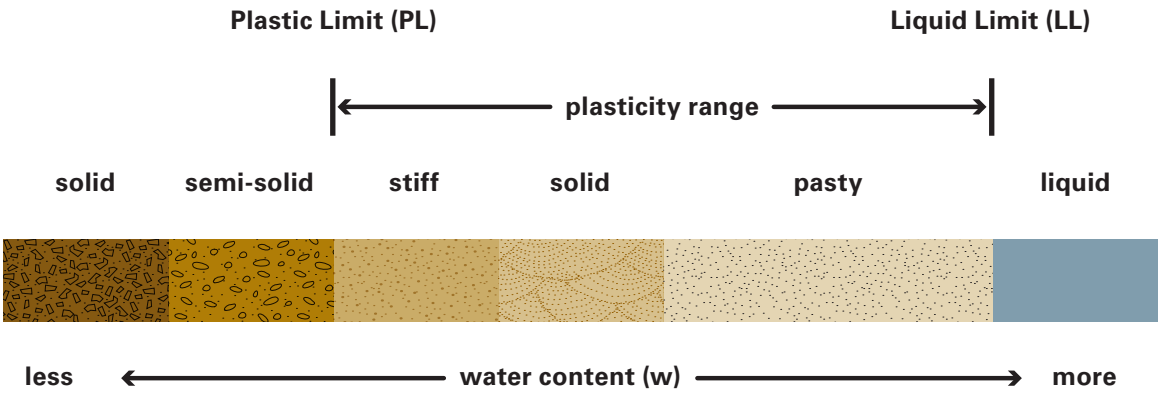
A simple test has been developed to determine the Liquid Limit of a soil. Take a moist sample of a soil and place it in a small bowl, flattening the sample somewhat. Make a deep groove in the sample and tap the bottom of the bowl 10-30 times, watching the groove. If the faces of the groove remain the same distance apart, pick up the sample, add more water, and repeat the process. When the faces of the groove move together on a length of 15 mm (½"), the sample has become somewhat liquid and has reached its Liquid Limit.

High LL values are associated with soils of high compressibility. Typically, clays have high LL values; sandy soils have low LL values.

Plastic Limit (PL)
This condition exists when a soil changes from a semi-solid to a plastic state. It occurs when the soil contains just enough moisture that a small amount of it can be rolled into a thread approximately 3 mm (1/8") in diameter without breaking.

The PL of a soil is important because it represents the moisture content at which particles will slide over each other and still possess appreciable cohesion. It is the point where best compaction occurs with high clay content soils. The strength of the soil decreases rapidly as the moisture content increases beyond the Plastic Limit.

PLASTICITY INDEX (PI)



Plasticity Index (PI)
This is the numerical difference between the Plastic Limit and Liquid Limit of a soil. Soils possessing high PI values are quite compressible and have high cohesion. Soil has little or no cohesion when the moisture content is at the Liquid Limit, but has considerable cohesion when the moisture content is at the Plastic Limit. Therefore, the PI offers a means of estimating the compressibility and cohesion of a soil.

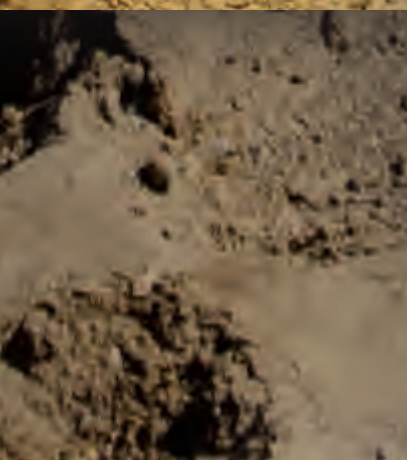
The PI also relates to permeability. The higher the PI, the lower the permeability—and the lower the PI, the higher the permeability. On many jobs involving construction with high clay content soils, the specifications call for material with a certain gradation, a maximum LL and a maximum PI.

Shrinkage Limit (SL)
As the soil is dried below the Plastic Limit, it shrinks and becomes brittle. At the moisture content that the sample volume change stops, the Shrinkage Limit is determined. The SL is the best moisture at which to compact many non-plastic (sandy) soils. Soils containing enough clay to raise the PI are best compacted somewhere between the SL and the PL.









Unit 2
SOIL TYPES AND CLASSIFICATIONS

Where possible, laboratory soil tests are the best option for classifying soils. When that is not possible, you can conduct one or more field tests to help you identify soils and determine a compaction approach.

[SOIL CLASSIFICATION SYSTEMS]

There are several different soil classification systems in use around the world today. All use the terms gravel, sand, silt and clay, but with slightly different numbering and lettering systems. The purpose of soil classifications is to create standards by which soils and their engineering characteristics can be identified.

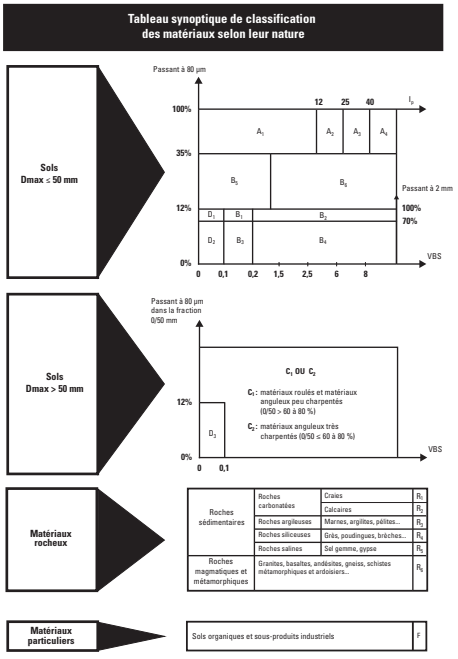
AASHTO Soil Classification System – The widely used American Association of State Highway and Transportation Officials (AASHTO) system of soil classification is based on field performance of soils for highway construction. The system divides materials into seven major groups with some subgroups. The groups are arranged in two major categories: granular materials and silty-clay materials.

AASHTO CLASSIFICATION OF HIGHWAY SUBGRADE MATERIALS
(with suggested subgroups)

General Classification	Granular Materials (35% or less passing #200)							Silt-Clay Materials (more than 35% passing #200)			
Group Classification	A-1		A-3	A-2				A-4	A-5	A-6	A-7 A-7-5 A-7-6
	A-1-a	A-1-b		A-2-4	A-2-5	A-2-6	A-2-7				
Sieve Analysis Percent Passing:											
# 10	0-50										
#40	0-30										
#200	0-15	0-50 0-25	51-100 0-10	0-35	0-35	0-35	0-35	36-100	36-100	36-100	36-100
Characteristics of Fraction Passing #40:											
Liquid Limit				0-40	41+	0-40	41+	0-40	41+	0-40	41+
Plasticity Index	0-6		N.P.	0-10	0-10	11+	11+	0-10	0-10	11+	11+
Group Index	0		0	0		0-4		0-8	0-12	0-16	0-20
Usual Types of Significant Constituent Materials	Stone Fragments Gravel and Sand		Fine Sand	Silty or Clayey Gravel and Sand				Silty Soils		Clayey Soils	
General Rating as Subgrade	Excellent to Good					Fair to Poor					

Please see Appendix for full size charts.

French Soil Classification System – This system classifies materials into classes and subclasses based on mechanical analysis of attributes including grain-size distribution, plasticity and sand equivalent.



German Soil Classification System – DIN18196 classifies all soil materials for construction purposes into groups based on particle size according to DIN4022, mass proportions, plasticity and the presence of organic and calcareous components. Generally, coarse particles and fine particles are judged differently as coarse particles have a particle-size distribution criterion and fines have a plasticity criterion.

Hauptgruppe	Kornanteil < 0,06 mm	Kornanteil > 2,0 mm	Gruppe (allgemein)	Gruppe (detailliert)	Kurzzeichen Gruppensymbol
Grobkörniger Boden	≤ 5	< 40	Kies	Enggestufte Kiese	GE
				Weitgestufte Kies-Sand-Gemische	GW
				Intermittierend gestufte Kies-Sand-Gemische	GI
			Sand	Enggestufte Sande	SE
				Weitgestufte Sand-Kies-Gemische	SW
				Intermittierend gestufte Sand-Kies-Gemische	SI
Gemischtkörniger Boden	5 bis 40	< 40	Kies-Schluff	5 bis 15 Gew.-% ≤ 0,06 mm	GU
				15 bis 40 Gew.-% ≤ 0,06 mm	GU*
			Kies-Ton	5 bis 15 Gew.-% ≤ 0,06 mm	GT
				15 bis 40 Gew.-% ≤ 0,06 mm	GT*
		≤ 40	Sand-Schluff	5 bis 15 Gew.-% ≤ 0,06 mm	SU
				15 bis 40 Gew.-% ≤ 0,06 mm	SU*
			Sand-Ton	5 bis 15 Gew.-% ≤ 0,06 mm	ST
				15 bis 40 Gew.-% ≤ 0,06 mm	ST*
Feinkörniger Boden	< 40	–	Schluff	Leicht plastische Schluffe W_L <35	UL
				Mittelplastische Schluffe W_L =35 bis 50	UM
			Ton	Leicht plastische Tone W_L ≤35	TL
				Mittelplastische Tone W_L =35 bis 50	TM
				Ausgeprägt plastische Tone W_L >50	TA
Organogener Boden	< 40	–	Nicht brenn- und schwelbar	Organogene Schluffe W_L =35 bis 50	OU
	≤ 40			Organogene Tone W_L >50	OT
				Grob bis gemischtkörnige Böden mit humosen Beimengungen	OH
				Grob bis gemischtkörnige Böden mit kalkigen, kieseligen Bildungen	OK
Organischer Boden	–	–	Brenn- und schwelbar	Nicht bis mäßig zersetzte Torfe	HN
				Zersetzte Torfe	HZ
				Mudden (Faulschlamm)	F
Auffüllung[1]	–	–	–	Auffüllung aus Fremdstoffen	A

1 - Eine Auffüllung ist eine unter menschlicher Einwirkung entstandene Schüttung aus natürlichen Böden oder Fremdstoffen.

Unified Soil Classification System – The USCS is a widely used method for classifying soils in construction projects. The U.S. Army Corps of Engineers and the U.S. Bureau of Reclamation developed this system. It uses texture as the descriptive term.

USCS SOIL CLASSIFICATION SYSTEM

SOIL FRACTION	SYMBOL	SIZE RANGE
Boulders	**None**	Greater than 12"
Cobbles	**None**	75 mm (3") to 12"
1- Course Grained Soils:		
Gravel	**G**	75 mm (3") to #4 Sieve (4.25 mm)
Course Gravel		75 mm to 19 mm
Fine Gravel		#4 Sieve to 19 mm
Sand	**S**	#4 Sieve to #200 Sieve (0.075 mm)
Course Sand		
Medium Sand		
Fine Sand		
2- Fine Grained Soils:		
Fines		Less than #200 Sieve
Silt	**M**	Use Atterberg Limits
Clay	**C**	Use Atterberg Limits
3- Organic Soils	**O**	**Use Atterberg Limits**
4- Peat	**Pt**	**Visual Identification**
Gradation Symbols		**Liquid Limit Symbols**
Well-graded	**W**	High LL **H**
Poorly-graded	**P**	Low LL **L**

United Kingdom Soil Classification System – British Standard (BS) system of classification is a protocol for identifying soil composition. Soil is first classified into either coarse or fine soil based on particle size. Granular soils are classified based on their particle-size distributions. Fine soils are sub-grouped based upon their plasticities.

SOIL GROUPS (GRAVEL and SAND may be qualified sandy GRAVEL and gravelly SAND where appropriate)			SUB-GROUPS and in-laboratory identification			
			GROUP SYMBOL	SUB-GROUP SYMBOL	FINES % < 0.06 mm	LIQUID LIMIT
COARSE SOILS less than 65% of material is finer than 0.06 mm	GRAVELS More than 50% of coarse material is of gravel size (greater than 2 mm)	Slightly silty or clayey GRAVEL	G	GW GP	GW GPu GPg	0 to 5	
		Silty GRAVEL	G-F	G-M	GWM GPM	5 to 15	
		Clayey GRAVEL		G-C	GWC GPC		
		Very silty GRAVEL	GF	GM	GML, etc.	15 to 35	
		Very clayey GRAVEL		GC	GCL GCI GCH GCV GCE		
	SANDS More than 50% of coarse material is of sand size (less than 2 mm)	Slightly silty or clayey SAND	S	SW SP	SW SPu SPg	0 to 5	
		Silty SAND	S-F	S-M	SWM SPM	15 to 35	
		Clayey SAND		S-C	SWC SPC		
		Very silty SAND	SF	SM	SML, etc.	15 to 35	
		Very clayey SAND		SC	SCL SCI SCH SCV SCE		
FINE SOILS more than 35% of material is finer than 0.06 mm	Gravelly or sandy SILTS and CLAYS 35% to 65% fines	Gravelly SILT	FG	MG	MLG, etc.		
		Gravelly CLAY		CG	CLG CIG CHG CVG CEG		< 35 35 to 70 50 to 70 70 to 90 > 90
		Sandy SILT	FS	MS	MLS, etc.		
		Sandy CLAY		CS	CLS, etc.		
	SILTS and CLAYS 65% to 100% fines	SILT (M SOIL)	F	M	ML, etc.		
		CLAY		C	CL CI CH CV CE		< 35 35 to 70 50 to 70 70 to 90 > 90
ORGANIC SOILS		Description letter "O" suffixed to say group or sub-group symbol			Organic matter in significant amount e.g. MHO – organic silt of high LL		
PEAT		Pt – consists predominantly of plant remains (fibrous or amorphous)					

Classification v1.00 Sept 2010

Primary Letter
G = Gravel
S = Sand
M = Silt
C = Clay
O = Organic Soil
Pt = Peat

Secondary Letter
W = Well graded
P = Poorly graded
M = With non-plastic fines
C = With plastic fines
L = Of low plasticity (LL < 50)
H = Of high plasticity (LL > 50)